

Question	Answer	Mark	Mark scheme	Additional guidance
18	Proof	M1 M1 C1 OR M1 M1 C1	begins proof to show that triangles ABO and CBO or triangles ABD and CBD are congruent by giving one pair of equal sides or equal angles with reason for different pair of equal sides or angles with reason for full proof that triangles ABO and CBO are congruent, SSS, or triangles ABD and CBD are congruent, RHS, and therefore angle $ABO = \text{angle } CBO$ $AB = CB$ (given) BO (or BD) is <u>common</u> $AO = CO$ <u>radii</u> of circle angle $BAD = \text{angle } BCD$ <u>angles</u> in a <u>semicircle</u> are 90 ($BO = AO = CO$ <u>radii</u> of circle) counts as two sides with reasons OR draws OA, OC and AC and labels angle $OAC = x$ and angle $OCA = x$ with reason given, $AO = CO$ <u>radii</u> of circle and base angles of an <u>isosceles triangle</u> are equal or $BAC = BCA$ since ABC is isosceles shows $OAC = OCA$ and shows $BAC = BCA$ and uses these to show $OAB = OCB$ with all reasons given for full proof concluding with angle $ABO = y$ and angle $CBO = y$ with reason given, eg $OA = OB = OC$ radii of circle and OBC and OAC are isosceles	Where D is point such that BOD is diameter